Problem Statement

Date	30 September 2026
Team ID	6ab4e2eba238dc999eb5fb00
Project Name	**Script-Controlled ACL – Restrict Record Access Based on Field Value**
Maximum Marks	3 Marks

Problem Statement

In modern organizations, ServiceNow is widely used to manage institutional information, user records, requests, and other business-related data. As the amount of information stored in a ServiceNow instance increases, controlling access to these records becomes an important security requirement. Providing users with unrestricted access to all records may expose information that is not relevant to their role, department, or organizational branch. Therefore, an effective access-control mechanism is required to ensure that users can access only the records they are authorized to view or modify.

The project **"Script-Controlled ACL – Restrict Record Access Based on Field Value"** addresses the problem of controlling access to records in a ServiceNow application based on the value of a particular field. In this project, an **Institution Details** table is created to store student-related information such as student number, student name, faculty, branch, email, phone number, and description. The Branch field contains multiple choices, including **ECE, EEE, and CSC**. Since the records belong to different branches, it is necessary to prevent users from accessing records belonging to branches for which they are not authorized.

The main problem is that role-based access control alone may not provide sufficient record-level restriction when access needs to depend on the value stored in a particular field. For example, assigning a role to a user may allow the user to access the Institution Details table, but the requirement is more specific: the user should be able to access only the records associated with the authorized branch. Without an additional condition, the user may be able to view records belonging to other branches as well. This can result in unnecessary exposure of institutional information and may violate the principle of least privilege.

To address this problem, the project uses **ServiceNow Access Control Lists (ACLs)** together with a **script-controlled authorization mechanism**. A user and multiple roles are created as part of the initial configuration. Four roles are created and assigned as required for controlling different operations. The Institution Details table is then created with appropriate fields and branch choices. Sample records are inserted into the table to represent different students and branches. These records provide the data required to verify whether the access-control mechanism is functioning correctly.

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The project specifically focuses on implementing ACLs for the four major database operations: **Read, Create, Write, and Delete**. A Read ACL is configured with a role and a branch-related condition. A script is incorporated into the ACL to provide additional authorization based on the value of the Branch field. Separate ACLs are then created for Create, Write, and Delete operations with the appropriate roles. This approach provides a structured mechanism for controlling what operations a user is permitted to perform on the Institution Details records.

The key challenge is to ensure that the access restrictions are applied correctly without preventing authorized users from accessing the records required for their work. The system must distinguish between authorized and unauthorized records based on the field value. For example, when the configured user accesses the Institution Details list, only records belonging to the permitted **EEE branch** should be displayed. Records associated with other branches should not be accessible to that user. At the same time, users with appropriate administrative privileges should continue to have the required access to the records.

Therefore, the problem addressed by this project is the need for a **secure, field-based record access mechanism in ServiceNow** that combines user roles, ACL rules, conditions, and scripting to restrict access at the record level. The proposed implementation provides a practical way to enforce authorization according to organizational requirements while maintaining separate permissions for Read, Create, Write, and Delete operations.

The successful implementation of this project demonstrates how ServiceNow ACLs can be used beyond simple role-based authorization to provide more controlled and field-dependent record security. The final system is expected to ensure that an authorized user can access the appropriate Institution Details records while unauthorized branch records remain restricted, thereby improving data security and supporting the principle of controlled access within the ServiceNow environment.